

Digital Twin / Jumeau numérique

What this tool produces · *A 3D, semantically rich digital replica of the city that integrates real-time sensor data, spatial geography, and infrastructure models to simulate urban processes.*

G2 · Proof of Fit

Dimensions

Spatial

Regulation

Economic

Preparing the workshop · *Digital Twin / Jumeau numérique*

G2 · Proof of Fit

Why this tool matters

Building a mirror world where the consequences of our political and architectural decisions can be explored safely, democratizing urban planning by making complex data visible to everyone.

Evidence we produce

A 3D, semantically rich digital replica of the city that integrates real-time sensor data, spatial geography, and infrastructure models to simulate urban processes.

Duration & Material

Duration · 1 to 3 years for city-scale platform development

Material · 3D city models, CityGML format, IoT sensor networks, cloud infrastructure, visualization platforms (e.g., Cesium)

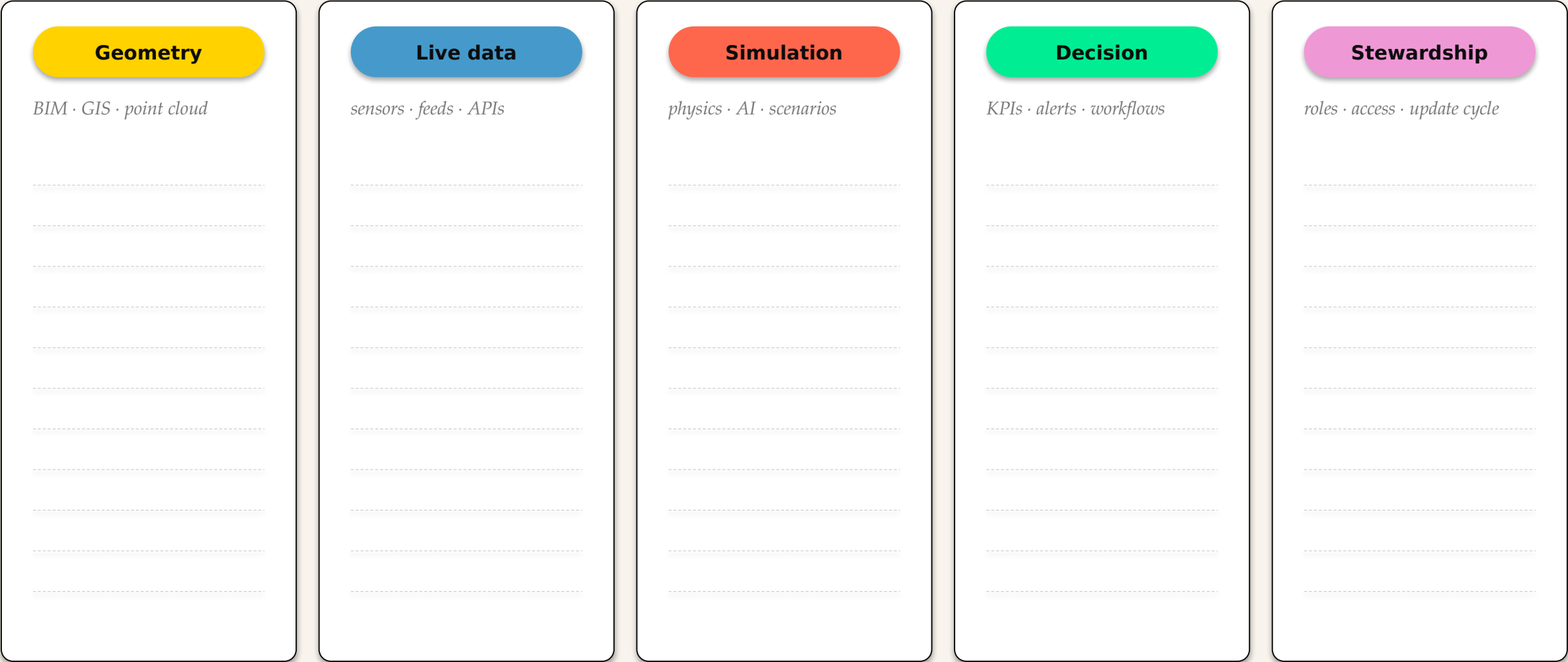
Reference case

The Kalasatama Digital Twins Project in Helsinki produced a high-quality CityGML information model shared as open data. This digital twin serves as a smart development platform for the entire lifecycle of the built environment. It allows the city, construction actors, and stakeholders to run simulations and collaboratively plan services before executing them in the physical world.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Digital twin · 5 layers integrated

Duration
1 to 3 years for city-scale platform development



A twin is only useful if it changes a real decision · scope to that.

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.